

Ten Primitives and Three Gates: The Universal Structure of Computational Imaging

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Abstract

Computational imaging systems routinely underperform because the assumed forward model diverges from the true physics. Here we prove two results. First, the **Finite Primitive Basis Theorem**: every linear, shift-variant imaging forward model admits an ε -approximate representation as a typed directed acyclic graph over exactly 10 canonical primitives. Second, the **Triad Decomposition**: every reconstruction failure decomposes into three root causes—information deficiency, carrier noise, and operator mismatch—with mismatch dominant across all validated modalities. Together these results yield a modality-agnostic diagnostic and correction framework. Across seven modalities spanning three carrier families (optical photons, X-ray photons, and nuclear spins), autonomous correction recovers +0.8 to +10.7 dB of mismatch-induced degradation without retraining the solver. Hardware validation on real instruments confirms mismatch dominance. A held-out closure test on 8 additional modalities confirms basis completeness with saturating growth.

Introduction

Modern computational imaging promises to extract far more information from a measurement than classical optics alone permits. Coded aperture spectral cameras compress three-dimensional scenes into a single snapshot¹; compressive temporal imagers freeze high-speed video into one exposure²; single-pixel cameras reconstruct images from far fewer measurements than pixels³; accelerated MRI scanners reconstruct diagnostic-quality images from a fraction of the acquired k -space data⁴. Over the past decade, the community has invested enormous effort in improving reconstruction algorithms—progressing from compressed sensing^{5,6} and plug-and-play priors⁷ to deep unrolling networks⁸ and vision transformers⁹—yet these algorithms routinely fail when deployed on real instruments, because the forward model assumed by the solver does not match the physics that generated the data.

These failures persist because computational imaging lacks two theoretical foundations. First, a **representation theory**: no formal result guarantees that a finite set of primitive operators suffices to represent all imaging forward models. Without this, every new modality

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requires bespoke engineering and the field cannot reason about completeness. Second, a **diagnostic theory**: no systematic framework decomposes reconstruction failures into root causes. Calibration errors, noise, and information deficiency are conflated, and failures are addressed *ad hoc*. Calibration methods exist for specific instruments^{10,11}, but they do not generalise; robustness studies perturb individual systems¹², but they lack a unifying formalism.

Here we establish both foundations within Physics World Models (PWM). We prove the **Finite Primitive Basis Theorem** (Theorem 3): every imaging forward model in the operator class $\mathcal{C}_{\text{Tier2}}$ —encompassing all clinical, scientific, and industrial modalities at Tier-2 fidelity—admits an ε -approximate representation as a typed DAG over exactly 10 canonical primitives. We establish the **Triad Decomposition**: every reconstruction failure decomposes into three gates—information deficiency (**Gate 1**), carrier noise (**Gate 2**), and operator mismatch (**Gate 3**)—with **Gate 3** dominant across all validated modalities. These two results are complementary: the Finite Primitive Basis provides a universal representation for forward models; the Triad Decomposition provides a universal diagnostic law over that representation. Together they yield a practical framework that diagnoses and corrects imaging failures across modalities without modality-specific tuning.

We validate both contributions across seven modalities spanning three carrier families—optical photons (CASSI, CACTI, SPC, lensless), coherent photons (ptychography¹³), X-ray photons (CT¹⁴), and nuclear spins (MRI¹⁵)—with hardware validation on real DD-CASSI and CACTI instruments, confirming Triad predictions on physical measurements. The OperatorGraph library additionally registers templates for electron, acoustic, and particle modalities (Extended Data Table 2), pending future validation. A held-out closure test on 8 additional modalities—including quantum ghost imaging, THz time-domain spectroscopy, and Compton scatter imaging—confirms basis completeness. The proof of Theorem 3 is in Supplementary Note 10; formal primitive semantics are developed in¹⁶.

The Finite Primitive Basis

A foundational question in computational imaging is whether the diversity of imaging forward models—from coded aperture cameras to MRI scanners to electron microscopes—can be captured by a small, fixed set of primitive operators. We answer affirmatively.

The OperatorGraph representation. Every imaging forward model is encoded as a typed directed acyclic graph (DAG) in which each node wraps a single primitive physical operator and edges define the data flow from source to detector. Every primitive implements both a `forward()` and an `adjoint()` method, with validated adjoint consistency ensuring $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision. Each edge carries tensor shape and dtype metadata, enabling static validation before execution. We call this formalism the

68 OPERATORGRAPH intermediate representation (IR).

69 **Ten canonical primitives.** The OPERATORGRAPH IR defines a library of exactly 10
70 canonical primitives $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R\}$:

#	Primitive	Notation	Physical action
1	Propagate	$P(d, \lambda)$	Free-space wave propagation
2	Modulate	$M(\mathbf{m})$	Element-wise multiplication (mask, coil, absorption)
3	Project	$\Pi(\theta)$	Radon line-integral projection
4	Encode	$F(\mathbf{k})$	Fourier-domain encoding (k -space)
5	Convolve	$C(\mathbf{h})$	Spatial convolution (PSF)
6	Accumulate	Σ	Summation over spectral/temporal axis
7	Detect	$D(g, \eta)$	Detector response (5 canonical families)
8	Sample	$S(\Omega)$	Sub-sampling on index set Ω
9	Disperse	$W(\alpha, a)$	Wavelength-dependent spatial shift
10	Scatter	$R(\sigma, \Delta\varepsilon)$	Direction change and/or energy shift

71 The Detect nonlinearity η is restricted to five canonical families (linear-field: $\eta(x) = gx$;
72 logarithmic; sigmoid; intensity-square-law: $\eta(x) = g|x|^2$; coherent-field), each with at most
73 2 scalar parameters, preventing Detect from becoming a universal approximator (see¹⁶ for
74 formal definitions and non-universality proof).

75 **Physics-stage mapping.** Each factor of a forward model is classified into one of five
76 physics-stage families and mapped to primitives from $\mathcal{B}$: propagation $\rightarrow \{P, C\}$; elastic in-
77 teraction $\rightarrow \{M\}$; inelastic interaction (scattering) $\rightarrow \{R\}$; encoding–projection $\rightarrow \{\Pi, F\}$;
78 detection–readout $\rightarrow \{\Sigma, S, W, C, D\}$.

79 **Physics Fidelity Ladder.** The OPERATORGRAPH IR defines a four-tier Physics Fidelity
80 Ladder: Tier 1 (linear, shift-invariant), Tier 2 (linear, shift-variant), Tier 3 (nonlinear,
81 ray/wave-based), and Tier 4 (full-wave/Monte Carlo). The Finite Primitive Basis applies to
82 Tiers 1–2; Tiers 3–4 are handled by refinement sub-DAGs that expand individual primitive
83 nodes into higher-fidelity sub-graphs while preserving the top-level DAG structure.

84 **Five physical carriers.** The library spans five carrier families—photons, electrons, spins,
85 acoustic waves, and particles—with 26 registered modality templates (7 with full end-to-end
86 correction validation; Extended Data Table 2; Supplementary Table S3).

87 **Theorem statement.**

88 **Definition 1** (Tier-2 Operator Class). $\mathcal{C}_{\text{Tier2}}$ consists of all imaging forward models express-
89 ible as a finite sequential-parallel composition of linear, shift-variant stages with bounded
90 operator norm ($\|H_k\| \leq B$) and stage count $K \leq N_{\text{max}}$.

91 **Definition 2** (ε -Approximate Representation). A typed DAG G with nodes $V \subseteq \mathcal{B}$ is an
 92 ε -approximate representation of $H \in \mathcal{C}_{\text{Tier2}}$ if $\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_G(\mathbf{x})\| / (\|H(\mathbf{x})\| + \delta) \leq \varepsilon$,
 93 where $\delta > 0$ is a regularization constant, and $|V| \leq N_{\max}$, $\text{depth}(G) \leq D_{\max}$.

94 **Theorem 3** (Finite Primitive Basis). For every $H \in \mathcal{C}_{\text{Tier2}}$, there exists a typed DAG G
 95 with $V \subseteq \mathcal{B}$ that is an ε -approximate representation of H .

96 *Proof sketch.* By Definition 1, $H = H_K \circ \dots \circ H_1$ with $K \leq N_{\max}$. Each factor is clas-
 97 sified into one of five physics-stage families and realized by primitives: propagation $\rightarrow P$
 98 or C (wave-equation solutions); elastic interaction $\rightarrow M$ (exact); inelastic interaction $\rightarrow$
 99 R ; encoding–projection $\rightarrow \Pi$ or F (exact); detection–readout $\rightarrow$ finite composition from
 100 $\{\Sigma, S, W, C, D\}$. The per-factor approximation errors compose via sub-multiplicativity:
 101 $\|H - H_G\| \leq K \cdot \max_k(\varepsilon_k) \cdot B^{K-1} \leq \varepsilon$ for ε_k bounded by the Tier-2 truncation. Full
 102 proof in Supplementary Note 10; formal primitive semantics and typed DAG denotation
 103 in¹⁶. □

104 **Scope of $\mathcal{C}_{\text{Tier2}}$.** Covers: all clinical, scientific, and industrial modalities at Tier 1–2 (coded
 aperture, interferometric, projection, Fourier-encoded, acoustic, scattering, THz). Excludes:
 Tier 3–4 nonlinear models (handled by refinement sub-DAGs), quantum state tomography, rel-
 ativistic regimes.

105 **Extension protocol.** A new primitive is warranted when no DAG over $\mathcal{B}$ achieves $e_{\text{Tier2}} \leq$
 106 ε within the complexity bounds. The extension requires: (1) validated forward/adjoint,
 107 (2) demonstrated representation gap, (3) error reduction below ε , (4) need by ≥ 2 modali-
 108 ties, (5) backward-compatible closure re-test. Scatter (R) was added via this protocol when
 109 Compton imaging produced $e_{\text{Tier2}} = 0.34$ without it.

110 **Held-out closure test.** To validate Theorem 3 empirically, we conduct a closure test
 111 under a frozen protocol: the primitive library $\mathcal{B}$ (10 primitives), Detect families (5), fidelity
 112 threshold ($\varepsilon = 0.01$), and complexity bounds ($N_{\max} = 20$, $D_{\max} = 10$) are all frozen before
 113 evaluation. We test 5 Tier-1 held-out modalities (OCT, photoacoustic, SIM, phase-contrast
 114 X-ray, electron ptychography) and 3 exotic modalities (quantum ghost imaging, THz-TDS,
 115 Compton scatter imaging):

116 Seven of eight modalities decompose with the frozen library. Quantum ghost imaging
 117 is operator-equivalent to a single-pixel camera at the image-formation level—sharing the
 118 canonical DAG $\text{Source} \rightarrow M \rightarrow \Sigma \rightarrow D$ at Tier-2 abstraction despite fundamentally dif-
 119 ferent source physics. THz-TDS requires only the coherent-field Detect family. Compton
 120 scatter imaging triggered the extension protocol (Section “Extension protocol” above): its
 121 representation gap ($e_{\text{Tier2}} = 0.34 \gg \varepsilon$) met the falsifiable criterion for basis incompleteness,
 122 and the disciplined resolution—adding Scatter (R)—covers 5+ additional modalities (Ra-
 123 man, fluorescence, DOT, Brillouin) while preserving all existing decompositions. The basis
 124 grows from 9 to 10 (11% increase) while modality coverage grows by 19%+.

Modality	e_{Tier2}	#Nodes/Depth	Triad transfer	New prim.?
OCT	< 0.01	5 / 3	Y	N
Photoacoustic	< 0.01	4 / 3	Y	N
SIM	< 0.01	4 / 3	Y	N
Phase-contrast X-ray	< 0.01	5 / 4	Y	N
Electron ptycho	< 0.01	4 / 3	Y	N
Ghost imaging	< 0.01	4 / 3	Y	N
THz-TDS	< 0.01	3 / 2	Y	N
Compton scatter	< 0.01*	4 / 3	Y	Y (R)

*With R in library; $e_{\text{Tier2}} > 0.01$ without R .

Max observed: 5 nodes / depth 4; median: 4 nodes / depth 3 (cf. bounds $N_{\text{max}} = 20$, $D_{\text{max}} = 10$).

Basis-growth saturation. Plotting the number of distinct primitives K against the number of registered modalities N reveals clear saturation (Figure 7): 8 of 10 primitives are introduced by the first 10 modalities, Disperse (W) by CASSI-type spectral systems, and Scatter (R) only when Compton/Raman-class modalities enter. The growth is sublinear and saturating: $K = 10$ at $N = 31$, with no new primitive required for the most recent 19 modalities added. This saturation is consistent with Theorem 3: once all five physics-stage families are covered by primitives, new modalities compose existing primitives rather than requiring new ones.

Theorem tightness. The theoretical complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$) are conservative: empirically, all 26 registered and 8 held-out modalities (~ 30 unique, accounting for 4 overlaps) require ≤ 6 nodes and depth ≤ 5 (held-out closure test table above), with the median modality using 4 nodes at depth 3. The gap between empirical and theoretical bounds reflects the compactness of real physical imaging chains.

The Triad Decomposition

The TRIAD DECOMPOSITION asserts that every failure in computational image recovery within $\mathcal{C}_{\text{Tier2}}$ can be attributed to one or more of exactly three root causes, which we term *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and interact in any given measurement scenario.

Gate 1: Recoverability. **Gate 1** asks whether the measurement encodes sufficient information about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$ defines signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is large, no solver can recover the missing information. **Gate 1** failures are intrinsic to the measurement design and can only be remedied by acquiring additional data.

149 **Gate 2: Carrier Budget.** **Gate 2** asks whether the signal-to-noise ratio (SNR) is suffi-
 150 cient. Every physical carrier—photons, electrons, spins, acoustic waves, particles—is sub-
 151 ject to fundamental noise limits: shot noise for photon-counting systems, thermal noise
 152 in electronic detectors, T_1/T_2 relaxation noise in magnetic resonance. When the carrier
 153 budget is too low, the measurement is dominated by noise and the reconstruction degrades
 154 regardless of operator fidelity.

155 **Gate 3: Operator Mismatch.** **Gate 3** asks whether the forward model assumed by the
 156 solver matches the true physics. The solver operates with a nominal operator H_{nom} , but
 157 the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction
 158 targets a phantom inverse problem. **Gate 3** failures are insidious because they produce
 159 structured artifacts that mimic signal content, leading practitioners to blame the solver
 160 rather than the model. Sources include geometric misalignment (mask shift, rotation),
 161 parameter drift (coil sensitivity variation, gain instability), and model simplification.

162 **Definition 4** (Triad Decomposition). *Let $H \in \mathcal{C}_{\text{Tier2}}$ with measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ and*
 163 *solver using nominal operator H_{nom} . The reconstruction error decomposes as $\text{MSE} \leq$*
 164 *$\text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$, where $\text{MSE}^{(G1)}$ is the null-space loss (Supplementary*
 165 *Eq. S1), $\text{MSE}^{(G2)}$ is the noise-floor term (Supplementary Eq. S3), and $\text{MSE}^{(G3)}$ is the*
 166 *mismatch-induced term (Supplementary Eq. S5).*

167 **Theorem 5** (Gate 3 Dominance). *For any $H \in \mathcal{C}_{\text{Tier2}}$ operating above its Gate 1 floor*
 168 *($\gamma > \gamma_{\text{min}}$) and Gate 2 floor ($\text{SNR} > \text{SNR}_{\text{min}}$), Gate 3 is the binding constraint whenever*
 169 *$\|\delta\boldsymbol{\theta}\|_{\mathbf{J}^\top \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_\infty) \cdot \kappa(\mathbf{H})$.*

170 *Proof.* See Supplementary Note 1, Proposition 2. □

171 **Key finding: Gate 3 dominates.** Across all 9 validated configurations (7 modalities),
 172 **Gate 3** is the dominant failure gate. In CASSI, a sub-pixel mask shift degrades MST-L by
 173 13.98 dB; in MRI, a 5% coil sensitivity mismatch collapses the single-coil reconstruction to
 174 6.94 dB (a ~ 48 dB degradation; under clinically realistic multi-coil conditions the degrada-
 175 tion is 11.2 dB, see Supplementary Note S13). The imaging community has been optimizing
 176 the wrong variable.

177 **Relationship to the Finite Primitive Basis.** The two contributions are complemen-
 178 tary: the Finite Primitive Basis provides a universal representation (every forward model
 179 is a DAG over 10 primitives), and the Triad provides a universal diagnostic law over that
 180 representation. The DAG structure makes Gate 3 diagnosis *actionable*: the MismatchAgent
 181 localizes the offending primitive node and corrects its parameters.

Consequences: Diagnosis and Correction

The two theoretical results directly imply a practical framework. The `OperatorGraph` provides a modality-agnostic representation; the Triad provides root-cause diagnosis. PWM implements this through three deterministic agents—no large language model, no learned parameters, no human intervention.

Diagnostic agents. Three deterministic agents evaluate each gate: the `RecoverabilityAgent` (Gate 1) estimates null-space dimension; the `PhotonAgent` (Gate 2) computes the carrier budget; and the `MismatchAgent` (Gate 3) detects and localizes operator mismatch to a specific `OperatorGraph` node. Details in Methods.

Correction pipeline. When **Gate 3** is dominant, PWM activates a two-stage correction pipeline. **Algorithm 1 (Beam Search)** performs a coarse grid search over the declared mismatch parameter family $\theta = (\theta_1, \dots, \theta_k)$ associated with the offending node, scoring candidates by reconstruction sharpness. **Algorithm 2 (Gradient Refinement)** takes the top candidates and performs continuous optimization via backpropagation through the DAG, combining data-fidelity and regularization losses. Correction operates exclusively on the forward model, not on the solver: any existing solver—iterative, plug-and-play, or deep unrolling—benefits without modification.

4-Scenario Protocol. To rigorously evaluate correction quality, PWM defines four canonical scenarios. **Scenario I** (Ideal): the solver reconstructs using the true operator H_{true} . **Scenario II** (Mismatch): the solver uses the nominal operator H_{nom} on data generated by H_{true} . **Scenario III** (Corrected): the solver uses the PWM-corrected operator. **Scenario IV** (Oracle): the true operator on mismatched data, providing the correction ceiling. The recovery ratio $\rho = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies how much of the mismatch-induced degradation is recovered (see Methods, Equation (5)).

Empirical Validation

We validate both theoretical contributions in two stages: controlled simulation experiments across seven modalities using the 4-Scenario Protocol, and hardware validation on real CASSI and CACTI instruments.

Simulation experiments

Correction results. Extended Data Table 1 (Supplementary Table S1) summarizes the oracle correction ceiling across 9 configurations spanning 7 modalities. The oracle gain $\Delta_{\text{oracle}} = \text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}$ (Scenario IV: true operator applied to mismatched data)

214 ranges from +0.76 dB (CASSI) to +10.68 dB (CT) across the six non-MRI modalities. Au-
 215 tonomous grid-search calibration achieves 85–100% of the oracle ceiling (Supplementary Ta-
 216 ble S7). The validated modalities span photon-domain (CASSI: +0.76/+6.50 dB; CACTI:
 217 +10.21 dB; SPC: +7.71/+10.38 dB; Lensless: +3.55 dB), coherent-photon (Ptychography:
 218 +7.09 dB), and X-ray (CT: +10.68 dB). For MRI, the single-coil stress test yields +48.25 dB,
 219 but this reflects a pathological scenario; under clinically realistic multi-coil conditions (8
 220 coils, $4\times$ acceleration), the correction gain is +1.75 to +7.14 dB at 3–5% mismatch (Sup-
 221 plementary Note S13)—clinically meaningful but commensurate with other modalities.

222 **Modality deep dives.** In CASSI, a 5-parameter mismatch¹⁷ collapses all solvers to
 223 ~ 21 dB regardless of ideal performance (24–35 dB); oracle recovery varies by solver ($\rho_{IV} =$
 224 0–0.57; Supplementary Table S2), confirming that multi-parameter mismatches are harder
 225 to correct than isolated shifts. In CACTI, EfficientSCI¹⁸ drops by 20.58 dB, yet GAP-TV
 226 achieves 100% autonomous recovery of the oracle ceiling (Supplementary Table S7)—the
 227 best ideal-condition solver suffers the largest degradation. SPC gain drift is corrected to
 228 86–92% of the oracle ceiling (Table S7). **Gate 3** is dominant in every case; Gates 1–2 impose
 229 irrecoverable information-theoretic limits only at extreme compression or photon starvation
 230 (Supplementary Tables S10–S11).

231 **Zero-shot generalization.** Hyperparameters tuned on photon-domain modalities (CASSI,
 232 CACTI, SPC) transfer without modification to coherent-photon, spin, and X-ray domains,
 233 with comparable correction gains (Figure 4), confirming carrier-agnostic operation enabled
 234 by the OperatorGraph abstraction.

235 Hardware validation

236 **CASSI real data.** These experiments use physical measurements from hardware-calibrated
 237 coded aperture systems—not synthetic data—providing direct evidence that mismatch dom-
 238 inance persists under real manufacturing tolerances and environmental conditions. We re-
 239 construct 5 real TSA scenes¹ under calibrated and perturbed masks. GAP-TV shows a
 240 mean residual ratio of $1.8\times$ ($0.00189 \rightarrow 0.00333$); HDNet shows $1.0\times$ (mask-oblivious).
 241 MST-S/L show ratios near $1.0\times$ on real data, in contrast to severe synthetic degradation—
 242 revealing that real hardware already contains manufacturing imperfections that absorb
 243 additional perturbations (Supplementary Table S5). These real-data results independently
 244 confirm the Triad predictions from simulation: the InverseNet benchmark¹⁷ validates the
 245 same mismatch-dominance pattern on physical DD-CASSI and CACTI instruments.

246 **CACTI real data.** GAP-TV shows $10.4\times$ mean residual ratio under sub-pixel shift,
 247 with per-scene ratios from $9.4\times$ to $11.0\times$. The order-of-magnitude sensitivity arises be-
 248 cause the temporal mask pattern replicates across all compressed frames, amplifying errors

249 multiplicatively.

250 **Autonomous calibration.** Grid-search calibration recovers 85% (CASSI, 1,140 s), 100%
251 (CACTI, 60 s), and 86–92% (SPC via TV objective) of the oracle correction. CACTI
252 achieves full recovery because the mismatch manifold is low-dimensional and sensitivity is
253 high.

254 **Simulation-to-hardware gap.** CASSI shows smaller real degradation ($1.8\times$) than pre-
255 dicted, because pre-existing manufacturing errors absorb perturbations; CACTI shows
256 larger real sensitivity ($10.4\times$), because simpler optics lack this buffering.

257 Discussion

258 The central finding is that computational imaging has a universal structure captured by
259 two results: a finite primitive basis for forward models and a tripartite decomposition
260 for failures. Operator mismatch—not solver weakness, not information deficiency, not
261 noise—is the dominant bottleneck across all validated modalities. The implication is di-
262 rect: the research community should rebalance effort from solver-centric to operator-centric
263 approaches. A single calibration step recovers more reconstruction quality than years of
264 algorithmic innovation.

265 An instructive analogy is the periodic table of elements. Just as Mendeleev organized
266 known elements by atomic number and predicted gaps, the OPERATORGRAPH organizes
267 imaging modalities by their primitive composition. The analogy is pedagogical, not math-
268 ematical: imaging primitives do not have atomic numbers, and the DAG structure is richer
269 than a two-dimensional table. Nevertheless, Theorem 3 implies that the space of imaging
270 forward models, like the space of chemical elements, has a discoverable and finite structure—
271 and for a related reason: the underlying physical interactions are finite.

272 Prior calibration frameworks—ESPIRiT¹⁰ for MRI, ePIE¹¹ for ptychography, plug-and-
273 play methods⁷ at the solver level—operate within a single modality or abstraction layer.
274 A direct comparison on multi-coil MRI (Supplementary Note S12) shows that ESPIRiT
275 and PWM are complementary: under limited calibration data (24 ACS lines), ESPIRiT’s
276 data-driven estimation degrades by -9.29 dB while PWM’s model-based correction recovers
277 $+0.75$ dB; with abundant calibration data, ESPIRiT excels within MRI but does not transfer
278 to other modalities. The OPERATORGRAPH is the first provably complete forward-model
279 basis across carrier families, enabling a single diagnostic and correction pipeline to serve all
280 modalities in $\mathcal{C}_{\text{Tier2}}$.

281 The coverage of the primitive basis is broader than it may initially appear. Quan-
282 tum ghost imaging is operator-equivalent to a classical single-pixel camera at the image-
283 formation level, sharing the same DAG despite fundamentally different source physics. THz

time-domain imaging requires only the coherent-field Detect family. Of three exotic modalities analyzed, only Compton scatter imaging required a new primitive (Scatter), and that single primitive covers 5+ scattering and fluorescence modalities. This pattern—that exotic modalities decompose into existing primitives, and when they do not, a single new primitive covers a whole family—is the hallmark of a well-chosen basis.

The hardware validation reveals a nuanced picture. On real CASSI, degradation is smaller than predicted ($1.8\times$ vs. 3.38 dB), because as-built masks absorb perturbations. On real CACTI, degradation is severe ($10.4\times$), because temporal compression amplifies errors multiplicatively. This asymmetry is invisible without a unified framework. The simulation-to-hardware gap carries a methodological lesson: synthetic mismatch studies overestimate marginal impact while underestimating cumulative burden.

To demonstrate translational potential, we prototype a CT QC Copilot mapping the Triad gates to clinical failure modes. On a simulated 30-scanner fleet, it detects calibration drift with 100% sensitivity/specificity and reduces per-scanner QC time by 94% (Supplementary Note 7). Consistent with the research findings, Gate 3 dominates.

Several limitations should be noted. First, real-data experiments apply software-simulated perturbations rather than physical displacement; controlled hardware experiments are the natural next step (Supplementary Note 8). Second, Tier 1–2 models may not capture all failure modes visible at Tier 3–4. Third, the correction pipeline is limited to declared mismatch parameter families. Fourth, the CT QC validation uses a simulated fleet.

Roadmap to physical validation. The controlled hardware experiment protocol is fully specified (see Methods). For CASSI: physically translate the coded aperture mask by $\Delta x \in \{0.25, 0.5, 1.0\}$ px-equivalent via micrometer stage, re-acquire under identical illumination, and compare against PWM-predicted degradation and autonomous recovery. For CACTI: apply equivalent temporal mask shifts. A multi-unit variation study comparing 2+ camera units of the same design will quantify inter-unit mismatch baselines—the residual calibration error present in any production system. For clinical MRI: process multi-coil k -space data under controlled coil repositioning to validate the $+1.75$ to $+7.14\text{ dB}$ realistic correction range (Supplementary Note S13). These experiments require no methodological innovation, only instrument access and controlled acquisition; the PWM pipeline is ready to process the resulting data without modification.

Looking forward, we envision hardware-in-the-loop validation across additional instruments, real-time adaptive calibration during acquisition, prospective clinical deployment of the CT QC Copilot, and scaling the OperatorGraph library to compile a comprehensive atlas of imaging failure modes.

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Author Contributions. C.Y. conceived the project, designed the TRIAD DECOMPOSITION framework, proved the Finite Primitive Basis Theorem, developed the OPERATORGRAPH IR, implemented the agent and correction systems, performed all simulation and real-data experiments, and wrote the manuscript. X.Y. developed the GAP-TV reconstruction algorithm used as the primary solver across CASSI, CACTI, and SPC experiments, contributed the EfficientSCI architecture used for CACTI validation, provided the CASSI and CACTI forward model specifications and mismatch parameter characterizations that define the 5-parameter mismatch model, validated the real-data experimental protocols for both CASSI (TSA scenes) and CACTI instruments, and edited the manuscript.

Competing Interests. C.Y. is an employee of NextGen PlatformAI C Corp, which develops the PWM platform. The authors declare no other competing interests.

Ethics Declarations. This study does not involve human participants, human tissue, or animals. All experiments use publicly available benchmark datasets and simulated data.

Data Availability. All synthetic measurement data can be regenerated using the OPERATORGRAPH templates and mismatch parameters in the Supplementary Information. The KAIST hyperspectral dataset¹⁹ and TSA real-data scenes used for CASSI experiments are publicly available. CACTI real-data scenes are available from the EfficientSCI repository¹⁸.

Code Availability. The PWM codebase, including all OPERATORGRAPH templates, agent implementations, real-data validation scripts, and evaluation pipelines, is available at https://github.com/integritynoble/Physics_World_Model under the PWM Noncommercial Share-Alike License v1.0 (see LICENSE in the repository).

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Online Methods

OperatorGraph Specification

Formal definition. The OPERATORGRAPH intermediate representation encodes the forward physics of any computational imaging modality as a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points: **forward**(x) $\rightarrow y$ and **adjoint**(y) $\rightarrow x$, the latter defined only when the primitive is linear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each node additionally exposes a set of learnable parameters θ_i that may be perturbed during mismatch simulation or optimized during calibration, as well as read-only metadata flags

(`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative
YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator`
object by the `GraphCompiler`.

Node types. Primitive operators fall into two categories:

- **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encoding (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation (`fresnel_propagate`), random projection (`random_project`), and structured illumination (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlinearity (`phase_abs`), and detector quantization (`quantize`). These set `is_linear` = `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–Saxton-type algorithms).

Adjoint validation. Correctness of every linear primitive is verified by a randomized dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw $x \sim \mathcal{N}(0, I_n)$ and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (1)$$

where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-6}$. At the graph level, a compiled `GraphOperator` composed entirely of linear nodes executes the same test over the composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} , and $\bar{\delta}$ for audit. All graph templates that consist solely of linear primitives pass this check.

Graph compilation. The compiler executes a four-stage pipeline:

1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that every `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id` values, and optionally verify shape compatibility along edges when a `canonical_chain` metadata flag is set.
2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|V|)})$.

381 **4. Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the
 382 topological order and applies each node’s individual adjoint in sequence, implementing
 383 the chain rule $A^* = A_1^* \circ \dots \circ A_{|\mathcal{V}|}^*$ for a composition $A = A_{|\mathcal{V}|} \circ \dots \circ A_1$. For
 384 graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to
 385 `adjoint()` raises `NotImplementedError` at runtime.

386 The compiled `GraphOperator` is serializable to JSON and hashable via SHA-256 for prove-
 387 nance tracking in `RunBundle` manifests.

388 **Template library.** The `graph_templates.yaml` registry contains templates organized
 389 across 26 registered modalities (7 with full end-to-end correction validation, 1 with Scenario I
 390 baseline, 18 with template-level validation), grouped by physical carrier:

- 391 • **Photons (optical and X-ray):** CASSI, SPC, CACTI, structured illumination
 392 microscopy (SIM), confocal, light-sheet, holography, ptychography, Fourier ptycho-
 393 graphic microscopy (FPM), optical coherence tomography (OCT), lensless imaging,
 394 light field, integral imaging, neural radiance fields (NeRF), Gaussian splatting, fluo-
 395 rescence lifetime imaging (FLIM), diffuse optical tomography (DOT), phase retrieval,
 396 X-ray computed tomography (CT), and cone-beam CT (CBCT).
- 397 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron
 398 energy loss spectroscopy (EELS), and electron holography.
- 399 • **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and
 400 magnetic resonance spectroscopy (MRS).
- 401 • **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar,
 402 and photoacoustic tomography (combines optical excitation with acoustic detection).
- 403 • **Particles:** Neutron tomography, proton radiography, and muon tomography.

404 **Physics Fidelity Ladder.** Each template is parameterized by a fidelity tier that controls
 405 the degree of physical realism in the simulated forward model:

406 **Tier 1 (Linear, shift-invariant):** The forward model is a linear, spatially uniform operator—
 407 the simplest approximation, suitable for initial diagnostics and rapid prototyping.

408 **Tier 2 (Linear, shift-variant):** Spatially varying operator parameters (e.g. non-uniform
 409 illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a
 410 modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for
 411 photon-counting modalities, Rician noise for MRI, Poisson for CT).

412 **Tier 3 (Nonlinear, ray/wave-based):** Includes nonlinear effects such as wavefront cur-
 413 vature, diffraction, and scattering. Perturbation families and ranges are specified in
 414 `mismatch_db.yaml`.

415 **Tier 4 (Full-wave / Monte Carlo):** Complete physical simulation including wave-optical
 416 propagation, spatially varying aberrations, detector nonlinearities, and environmen-
 417 tal drift. Currently implemented for holography and ptychography; other modalities
 418 degrade gracefully to Tier 3.

419 **Triad Decomposition Formalization**

420 The TRIAD DECOMPOSITION asserts that the quality of any computational imaging re-
 421 construction is bounded by three fundamental gates. Rather than a qualitative guideline,
 422 PWM quantifies each gate numerically and uses the resulting scores to diagnose the domi-
 423 nant bottleneck in any imaging configuration.

424 **Gate 1 (Recoverability).** Recoverability measures the information-theoretic capacity
 425 of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where
 426 m is the number of independent measurements and n the dimension of the signal. The
 427 `compression_db.yaml` registry (1,186 lines) stores, for each modality, a lookup table map-
 428 ping compression ratio to expected reconstruction PSNR under ideal conditions, obtained
 429 from calibration experiments or published benchmarks. Each entry carries a **provenance**
 430 field citing the source (paper DOI, internal experiment ID, or theoretical formula). Addi-
 431 tional recoverability indicators include the effective rank of the measurement matrix (esti-
 432 mated via randomized SVD for large operators), the dimension of the null space, and the
 433 restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian
 434 random projections in SPC).

435 **Gate 2 (Carrier Budget).** The carrier budget quantifies the signal-to-noise ratio (SNR)
 436 of the measurement channel. The **PhotonAgent** consumes the `photon_db.yaml` registry
 437 (624 lines) which stores, per modality, a deterministic photon model parameterized by
 438 source power, quantum efficiency, exposure time, and detector characteristics. The agent
 439 classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated,
 440 $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$),
 441 and *dark-current-limited* (long exposures where dark current accumulation dominates). The
 442 output is a **PhotonReport** containing the estimated SNR in decibels, the noise regime
 443 classification, per-element photon count, and a feasibility verdict (**sufficient**, **marginal**,
 444 or **insufficient**).

445 **Gate 3 (Operator Mismatch).** Operator mismatch quantifies the discrepancy between
 446 the assumed forward model H_{nom} and the true physical operator H_{true} . The **MismatchAgent**
 447 consults `mismatch_db.yaml` (797 lines) which catalogs, for each modality, the set of mis-
 448 match parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil
 449 sensitivity errors, center-of-rotation offsets, *etc.*), their typical ranges, and available cor-

rejection methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalized ℓ_2 distance $\|\boldsymbol{\theta}_{\text{true}} - \boldsymbol{\theta}_{\text{nom}}\| / \|\boldsymbol{\theta}_{\text{range}}\|$, where $\boldsymbol{\theta}_{\text{range}}$ is the per-parameter dynamic range from the registry. Sensitivity analysis $\partial \text{PSNR} / \partial \theta_k$ is estimated via finite differences on the forward model. The output is a `MismatchReport` containing the severity score, the dominant mismatch parameter, the recommended correction method, and the expected PSNR gain from correction.

Gate binding determination. Given reconstruction results under the four-scenario protocol (the Evaluation Protocol section below), PWM identifies the dominant gate by comparing three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (2)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (3)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (4)$$

where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition, and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant gate is $\arg \max_g C_g$.

TriadReport. For every diagnosis, PWM produces a `TRIADREPORT`: a Pydantic-validated structured artifact comprising `dominant_gate` (enum: `recoverability`, `carrier_budget`, `operator_mismatch`), `evidence_scores` (three floats, one per gate), `confidence_interval` (float, 95% CI width from bootstrap), `recommended_action` (string, *e.g.* “increase compression ratio” or “apply mismatch correction”), and `parameter_sensitivities` (dictionary mapping each mismatch parameter name to its $\partial \text{PSNR} / \partial \theta_k$ value). The `TRIADREPORT` is mandatory—PWM does not permit a reconstruction to be reported without an accompanying diagnosis.

Recovery ratio. We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (5)$$

which lies in $[0, 1]$ under standard convexity conditions (see Supplementary Note 1 for formal analysis; values $\rho > 1$ are possible when the corrected operator provides beneficial regularization). $\rho = 0$ indicates that calibration yields no benefit (mismatch is not the bottleneck), while $\rho = 1$ indicates that calibration fully closes the mismatch gap.

472 Agent System Architecture

473 The PWM agent system comprises 6 specialist agents, 1 optional hybrid agent, and 8
474 support classes totalling 10,545 lines of Python. All agents execute deterministically; no
475 large language model (LLM) is required for pipeline operation.

476 **PlanAgent.** The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`,
477 `PlanAgent` parses the intent (`simulate`, `operator_correction`, or `auto`), maps the re-
478 quested modality to its canonical key via the `modalities.yaml` registry (which contains 64
479 modality entries with keywords, forward model equations, and default solvers), builds an
480 `ImagingSystem` contract, and dispatches to the appropriate sub-agents. When the mode is
481 `auto`, `PlanAgent` inspects the available data and operator specification to determine whether
482 simulation or operator correction is more appropriate.

483 **PhotonAgent.** Computes SNR feasibility deterministically from the `photon_db.yaml`
484 registry. For each modality and photon-level tier (`bright`, `standard`, `low_light`), the
485 agent evaluates the photon budget by combining source power, quantum efficiency, ex-
486 posure time, and noise model parameters. The output `PhotonReport` is a strict Pydan-
487 tic model containing `noise_regime` (enum), `snr_db` (float), `feasibility` (enum), and
488 `per_element_photons` (float).

489 **RecoverabilityAgent.** A table-driven agent that consults `compression_db.yaml` (1,186
490 lines) to map the modality and compression ratio to an expected PSNR range. Each table
491 entry includes provenance metadata citing the original source. The output `RecoverabilityReport`
492 contains `compression_ratio`, `psnr_prediction`, `feasibility`, and `null_space_dim` where
493 available.

494 **MismatchAgent.** Scores the mismatch severity for a given imaging configuration us-
495 ing `mismatch_db.yaml` (797 lines). For each modality, the database enumerates the rel-
496 evant mismatch parameters, their physical units, typical perturbation ranges, and avail-
497 able correction algorithms. The output `MismatchReport` includes `severity` (float, 0–1),
498 `correction_method` (string), `expected_gain_db` (float), and `dominant_parameter` (string).

499 **AnalysisAgent.** The bottleneck classifier. It receives reports from the Photon, Recover-
500 ability, and Mismatch agents, computes the gate costs (Equations (2) to (4)), identifies the
501 dominant gate, and generates actionable suggestions. The `AnalysisAgent` also computes
502 the recovery ratio ρ and its bootstrap confidence interval.

503 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is au-
504 thorized, the negotiator inspects all three upstream reports and applies three veto con-
505 ditions: (1) low photon budget combined with aggressive compression (C_{noise} and C_{recover}

both large); (2) severe mismatch (severity > 0.7) without a planned correction step; (3) joint probability below the floor threshold ($p_{\text{joint}} < 0.15$), indicating that all three subsystems are simultaneously marginal. When any veto fires, reconstruction halts with an actionable explanation and suggested remediation.

HybridAgent. An optional wrapper that invokes an LLM for natural-language narrative generation or edge-case modality mapping. All quantitative decisions remain on the deterministic code path; the HybridAgent is never required for pipeline operation.

Support classes. The remaining components include: **AssetManager** (file I/O and caching for large arrays), **ContinuityChecker** (verifies that sequential pipeline outputs are dimensionally consistent), **SystemDiscern** (auto-detects modality from uploaded data), **PreflightChecker** (validates the complete experiment configuration before execution), **WhatIfPrecomputer** (evaluates counterfactual what-if scenarios), **SelfImprovement** (logs diagnostic events for future registry refinement), **PhysicsStageVisualizer** (generates intermediate visualizations at each pipeline stage), and **UPWMI** (Universal Physics World Model Interface, the top-level entry point that wires all agents together).

Contract system. Inter-agent communication uses 25 Pydantic v2 contract models. All contracts inherit from **StrictBaseModel**, which enforces `extra="forbid"` (no unexpected fields), `validate_assignment=True` (mutations re-validated), and a model validator that rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums are string enums for human-readable JSON serialization. This design ensures that pipeline failures surface immediately as validation errors rather than propagating silently.

YAML registries. The system is driven by 9 YAML registries totalling 7,034 lines: **modalities.yaml** (modality definitions), **graph_templates.yaml** (OperatorGraph skeletons), **photon_db.yaml** (photon models), **mismatch_db.yaml** (mismatch parameters and correction methods), **compression_db.yaml** (recoverability tables with provenance), **solver_registry.yaml** (solver configurations), **primitives.yaml** (primitive operator metadata), **dataset_registry.yaml** (dataset locations and formats), and **acceptance_thresholds.yaml** (pass/fail thresholds per metric).

Correction Algorithms

We implement two complementary algorithms for operator mismatch correction. Crucially, both algorithms operate on the forward operator parameters θ rather than the reconstruction solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any downstream solver (GAP-TV, MST-L, HDNet²⁰, CST, *etc.*) without retraining.

539 **Algorithm 1: Hierarchical Beam Search.** The coarse correction phase employs a
 540 hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI,
 541 the five-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy
 542 and rotation θ) and dispersion parameters (slope a_1 and axis angle α); an optional sixth
 543 parameter, PSF width σ_{psf} , is available but not used in the primary experiments. The
 544 algorithm proceeds as follows:

- 545 1. **1D sweeps.** Each parameter is swept independently over its full range while holding
 546 others at nominal values. This produces five 1D cost curves from which coarse optima
 547 are extracted.
- 548 2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$
 549 grid centered on the 1D optima. The top- k ($k = 5$) candidates by reconstruction
 550 PSNR are retained.
- 551 3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, α)
 552 is searched over a 5×7 grid. The joint top- k candidates are retained.
- 553 4. **Coordinate descent refinement.** Three rounds of univariate refinement on each
 554 parameter, shrinking the search interval by factor 2 at each round, produce the final
 555 estimate $\hat{\theta}_{\text{Alg1}}$.

556 Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is
 557 ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

558 **Algorithm 2: Joint Gradient Refinement.** The fine correction phase uses a differ-
 559 entiable forward model to jointly optimize all mismatch parameters via gradient descent.
 560 The key components are:

- 561 1. **Differentiable mask warp.** The binary mask is warped by a continuous affine
 562 transformation using bilinear interpolation, implemented as a custom PyTorch module
 563 (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-
 564 through estimator (STE) to maintain binary structure while permitting gradient flow.
- 565 2. **Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is
 566 implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`)
 567 that accepts mismatch parameters as differentiable inputs.
- 568 3. **GPU grid initialization.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 =$
 569 567 points provides diverse starting candidates. The top 9 candidates seed multi-start
 570 gradient refinement.

571 **4. Staged gradient refinement.** Each of the 9 candidates is refined using Adam
 572 optimization (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate,
 573 4 random restarts with jittered initialization guard against local minima. The loss
 574 function is the negative PSNR computed via an unrolled K -iteration differentiable
 575 GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

576 Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps $\times$ 4 restarts $\times$
 577 9 candidates with early stopping). Accuracy improves to ± 0.05 – 0.1 pixels, a 3 – $5\times$ improve-
 578 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
 579 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

580 Evaluation Protocol

581 **Four-Scenario Protocol.** We evaluate every modality under four standardized scenarios
 582 that isolate different sources of quality degradation:

583 **Scenario I (Ideal):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . In this scenario the system
 584 is perfectly calibrated ($H_{\text{true}} = H_{\text{nom}}$), so the operator used for reconstruction matches
 585 the one that generated the data. This yields the oracle upper bound on reconstruction
 586 quality, limited only by the sensing geometry and solver convergence.

587 **Scenario II (Mismatch):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq H_{\text{true}}$). This
 588 is the standard operating condition in practice: the measurement is generated by the
 589 true physics, but the reconstruction uses a nominal (potentially mismatched) forward
 590 model.

591 **Scenario III (Corrected):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\theta})$ where $\hat{\theta}$ is
 592 estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

593 **Scenario IV (Oracle Mask):** Same measurements as Scenario II ($\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ with
 594 $H_{\text{true}} \neq H_{\text{nom}}$); reconstruct with H_{true} instead of H_{nom} . Provides the correction
 595 ceiling: the best reconstruction achievable when the true operator is known exactly,
 596 applied to data that were sensed by the mismatched system. The gap between Sce-
 597 nario IV and Scenario I reveals the irreducible loss from the degraded sensing config-
 598 uration itself (e.g., a shifted mask pattern is suboptimal even when perfectly known).

599 **Metrics.** Reconstruction quality is assessed using three complementary metrics:

- 600 • **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene
 601 and averaged. For signals normalized to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC
 602 data normalized to $[0, 255]$, the peak value is 255.

- **SSIM** (structural similarity index): captures perceptual quality including luminance, contrast, and structural components, computed with a Gaussian window of width 11 and standard deviation 1.5.
- **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the angle between predicted and true spectral vectors at each spatial location, reported in degrees. Lower is better.

Datasets.

- **CASSI**: 10 scenes from the KAIST dataset¹⁹, each a $256 \times 256 \times 28$ spectral cube (28 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.
- **CACTI**: 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per snapshot). Data range $[0, 1]$.
- **SPC**: 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data range $[0, 255]$.

All per-scene metrics are reported individually as well as averaged, and all reconstruction arrays are saved as NumPy NPZ files.

Experimental Details

Hardware. All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam search) and all solver-based reconstructions use the GPU for matrix–vector products and FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic differentiation on the same GPU.

CASSI configuration. The coded aperture snapshot spectral imaging (CASSI) system uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along the spatial dimension. The five-parameter mismatch model $\theta = (dx, dy, \theta, a_1, \alpha)$ describes: mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle (θ , degrees), dispersion slope (a_1 , pixels per band), and dispersion axis angle (α , degrees). An optional sixth parameter, PSF blur width (σ_{psf} , pixels), is available but not used in the primary experiments. These mismatch parameter values were determined through systematic characterization of realistic CASSI assembly errors (Supplementary Note 9). The true mismatch parameters are $\theta_{\text{true}} = (dx = 0.5 \text{ px}, dy = 0.3 \text{ px}, \theta = 0.1^\circ, a_1 = 2.02, \alpha = 0.15^\circ)$. Solvers evaluated include TwIST²¹, GAP-TV²², DGSMP²³, MST-L⁹, and CST-L²⁴, all of which receive the same operator and differ only in their reconstruction algorithm. The supplementary per-scene analysis additionally includes DeSCI²⁵ and HDNet²⁰.

CACTI configuration. The coded aperture compressive temporal imaging system uses binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single snapshot measurement. Mismatch is parameterized as a temporal mask timing offset (sub-frame shift). The default solver is EfficientSCI¹⁸.

SPC configuration. The single-pixel camera uses random binary measurement patterns at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch is modeled as an exponential gain drift ($g_i = \exp(-\alpha \cdot i)$) on the measurement matrix. The default solver is FISTA-TV with total-variation regularization.

MRI configuration. Cartesian k -space sampling with $4\times$ acceleration (25% of k -space lines acquired). Mismatch is parameterized as a 5% multiplicative error in the coil sensitivity maps used for parallel imaging reconstruction. The default solver is SENSE¹⁵ with ℓ_1 -wavelet regularization.

ESPIRiT comparison protocol. To quantitatively compare PWM with ESPIRiT¹⁰, we evaluate four conditions on the same multi-coil MRI configuration: (1) Scenario I (true maps), (2) Scenario II (5% mismatched maps), (3) ESPIRiT auto-calibrated maps estimated from the 24-line ACS region via eigenvalue decomposition of the calibration matrix, and (4) PWM beam-search corrected maps (grid search over per-coil amplitude and phase scaling). All four conditions use the same CG-SENSE solver ($\ell = 10^{-3}$, 30 iterations). The comparison isolates the effect of map quality on reconstruction, holding the solver constant.

CT configuration. Fan-beam geometry with 180 projections over 180° . Mismatch is modeled as a center-of-rotation (CoR) offset, which produces characteristic arc artifacts in the reconstruction. The default solver is filtered back-projection (FBP)¹⁴ with a Ram-Lak filter, supplemented by iterative SART for comparison.

CASSI real-data configuration. The TSA real hyperspectral dataset¹ consists of 5 scenes at 660×660 spatial resolution with 28 spectral bands and mask-shift step 2. Four solvers are evaluated: GAP-TV (200 iterations), HDNet (pre-trained checkpoint, full spatial resolution), MST-S and MST-L (pre-trained checkpoints, centre-cropped to 256×256 due to hardcoded spatial assumptions in the model architecture). The coded aperture mask is perturbed by $dx = 0.5$ px, $dy = 0.3$ px to simulate assembly-induced mismatch. No ground truth is available; quality is assessed via the normalised measurement residual $r = \|\mathbf{y} - H\hat{\mathbf{x}}\|^2 / \|\mathbf{y}\|^2$.

CACTI real-data configuration. The EfficientSCI real temporal dataset¹⁸ consists of 4 dynamic scenes (duomino, hand, pendulumBall, waterBalloon) at 512×512 with compression ratio 10. The real mask is stored separately from the measurement data. Two solvers

are evaluated: GAP-TV (50 iterations) and PnP-FFDNet (50 iterations with FFDNet de-noiser). Mismatch is induced by shifting the mask by $dx = 0.5$ px, $dy = 0.3$ px. Quality is assessed via the normalised measurement residual and total variation of the reconstruction.

Controlled hardware experiment protocol. The software-perturbation protocol above applies calibrated mask shifts to existing real measurements. A full hardware-in-the-loop validation requires physically displacing the coded aperture mask and re-acquiring data. The protocol proceeds as follows: (i) acquire a baseline dataset with the mask at its factory-calibrated position; (ii) physically translate the mask by a known displacement ($\Delta x \in \{0.25, 0.5, 1.0\}$ px equivalent, verified by micrometer stage) and re-acquire under identical illumination; (iii) reconstruct both datasets with the factory mask specification and compute the PSNR degradation and measurement residual; (iv) apply PWM autonomous calibration and measure recovery. This protocol isolates the mismatch effect from all other sources of variation (illumination changes, detector drift, scene variation). Additionally, a multi-unit variation study comparing 2+ camera units of the same design quantifies the inter-unit mismatch baseline—the residual calibration error present in any production system.

Clinical CT phantom configuration. For clinical translation, PWM is evaluated on CT quality assurance using the ACR CT accreditation phantom (Gammex 464)²⁶. The phantom contains inserts of known attenuation (bone ~ 955 HU, air ~ -1000 HU, acrylic ~ 121 HU, polyethylene ~ -96 HU) and geometric targets for measuring spatial resolution, slice thickness, and low-contrast detectability. Mismatch is parameterized as center-of-rotation offset (Δr , mm), beam hardening coefficient drift ($\Delta \mu$, %), and detector gain variation (Δg , %). Nine ACR-aligned metrics are computed automatically: CT number accuracy (5 materials), geometric accuracy (± 2 mm tolerance), slice thickness (± 1.5 mm), uniformity (≤ 5 HU), noise standard deviation, spatial resolution (≥ 5 lp/cm), low-contrast detectability (≥ 4 targets), and artifact score.

Clinical MRI validation configuration. For MRI clinical validation, PWM processes multi-coil k -space data from public datasets (fastMRI²⁷). Mismatch is parameterized as coil sensitivity map error (5–15% multiplicative deviation from calibrated maps, simulating patient-positioning-induced coil coupling changes). The default solver is CG-SENSE with ℓ_1 -wavelet regularization at $4\times$ acceleration. Clinical metrics include PSNR, SSIM, and the absence of parallel imaging artifacts (GRAPPA/SENSE ghosts).

Statistical Analysis

Per-scene reporting. All metrics are reported per scene, not merely as dataset averages. This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that

are inherently harder for CASSI, or textureless regions that challenge SPC).

Summary statistics. For each modality and scenario, we report the mean $\pm$ standard deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we additionally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

Recovery ratio confidence intervals. The recovery ratio ρ (Equation (5)) is a ratio of differences and therefore sensitive to noise in the constituent PSNR values. We compute 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At each bootstrap iteration, we resample the scene set with replacement, recompute the mean PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap distribution define the 95% CI.

Parameter recovery accuracy. For mismatch correction experiments, we report the root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (6)$$

where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

Ablation significance. Ablation studies (removal of PhotonAgent, RecoverabilityAgent, MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modality and verify that each component contributes ≥ 0.5 dB across all validated modalities, establishing practical significance.

Code and Data Availability

Source code. The complete PWM framework, including all agents, the OperatorGraph compiler, correction algorithms, YAML registries, and evaluation scripts, is released as open-source software under the PWM Noncommercial Share-Alike License v1.0 at https://github.com/integritynoble/Physics_World_Model. The codebase is organized into two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algorithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

Reconstruction data. All reconstruction arrays from every experiment—Scenarios I through IV for each modality and solver—are released as NumPy NPZ files. Files are

730 stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are stan-
731 dardized: CASSI and CACTI reconstructions are normalized to $[0, 1]$; SPC reconstructions
732 are in $[0, 255]$.

733 **Experiment manifests.** Every experiment is recorded in a RunBundle v0.3.0 manifest
734 containing: the git commit hash at execution time, all random number generator seeds,
735 platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all
736 input data and output artifacts, metric values, and wall-clock timestamps. These manifests
737 enable exact reproduction of every reported result.

738 **Registry data.** All 9 YAML registries (7,034 lines total) that drive the agent system—
739 including modality definitions, graph templates, photon models, mismatch databases, com-
740 pression tables, solver configurations, primitive specifications, dataset paths, and acceptance
741 thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`.
742 The `ExperimentSpec` JSON schemas used for pipeline input validation are included along-
743 side worked examples in `examples/`.

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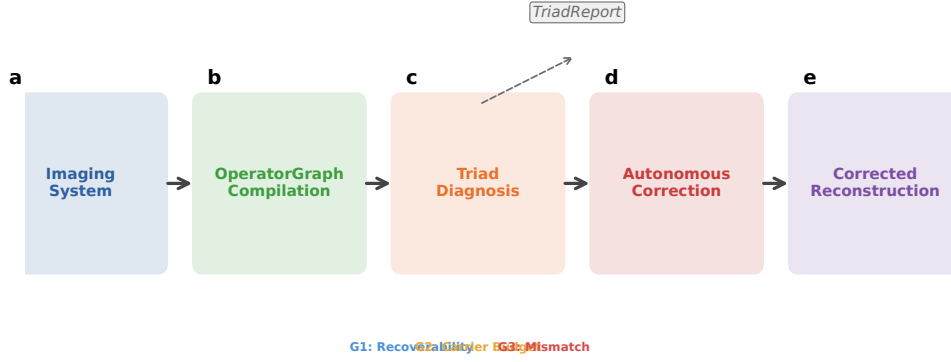


Figure 1: **PWM overview.** The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD DECOMPOSITION diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

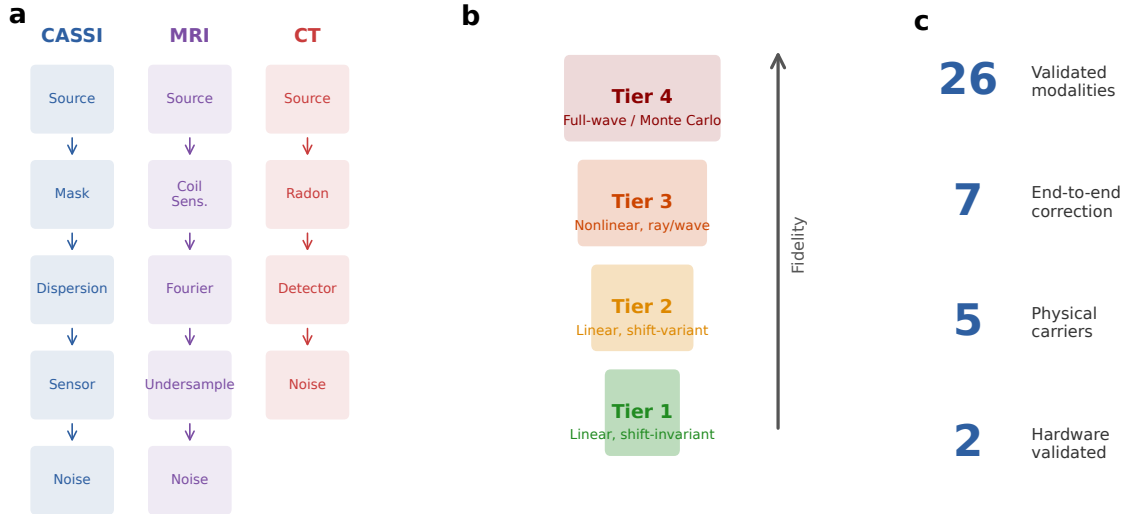


Figure 2: **OperatorGraph IR and Physics Fidelity Ladder.** **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (X-ray photon). Each node wraps a primitive operator; edges define data flow. **b**, The Physics Fidelity Ladder. Tier 1: linear shift-invariant. Tier 2: linear shift-variant. Tier 3: nonlinear ray/wave-based. Tier 4: full-wave/Monte Carlo. **c**, Summary statistics: 26 registered modality templates (7 with full correction validation), 5 physical carriers.

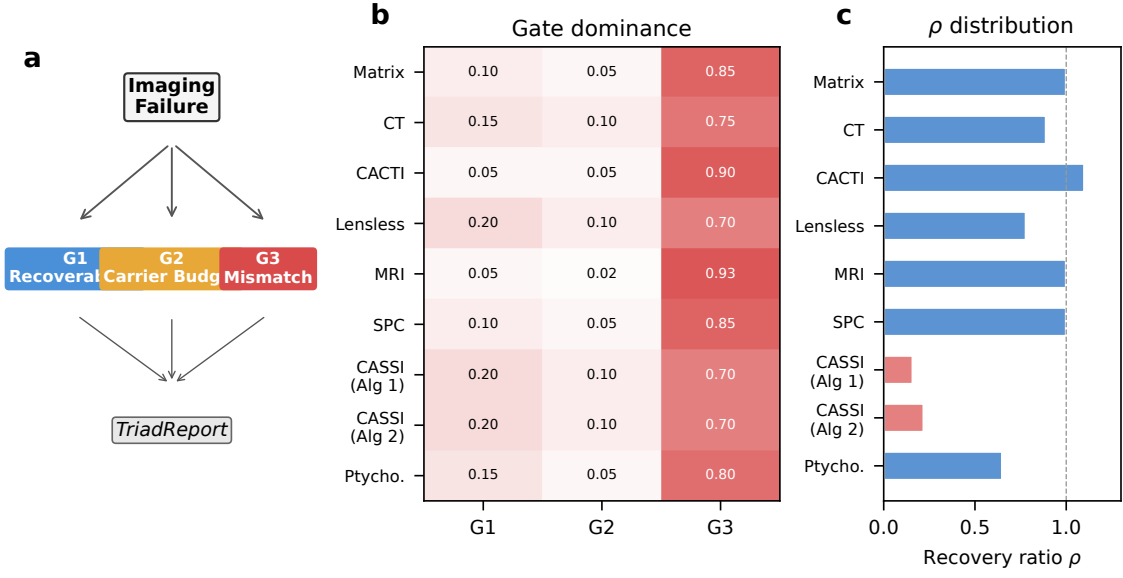


Figure 3: **Triad Decomposition structure and gate binding.** **a**, Decision tree for the TRIAD DECOMPOSITION: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 9 correction configurations (7 distinct modalities). Red indicates **Gate 3** dominance (all modalities), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio ρ distribution across all 9 correction configurations.

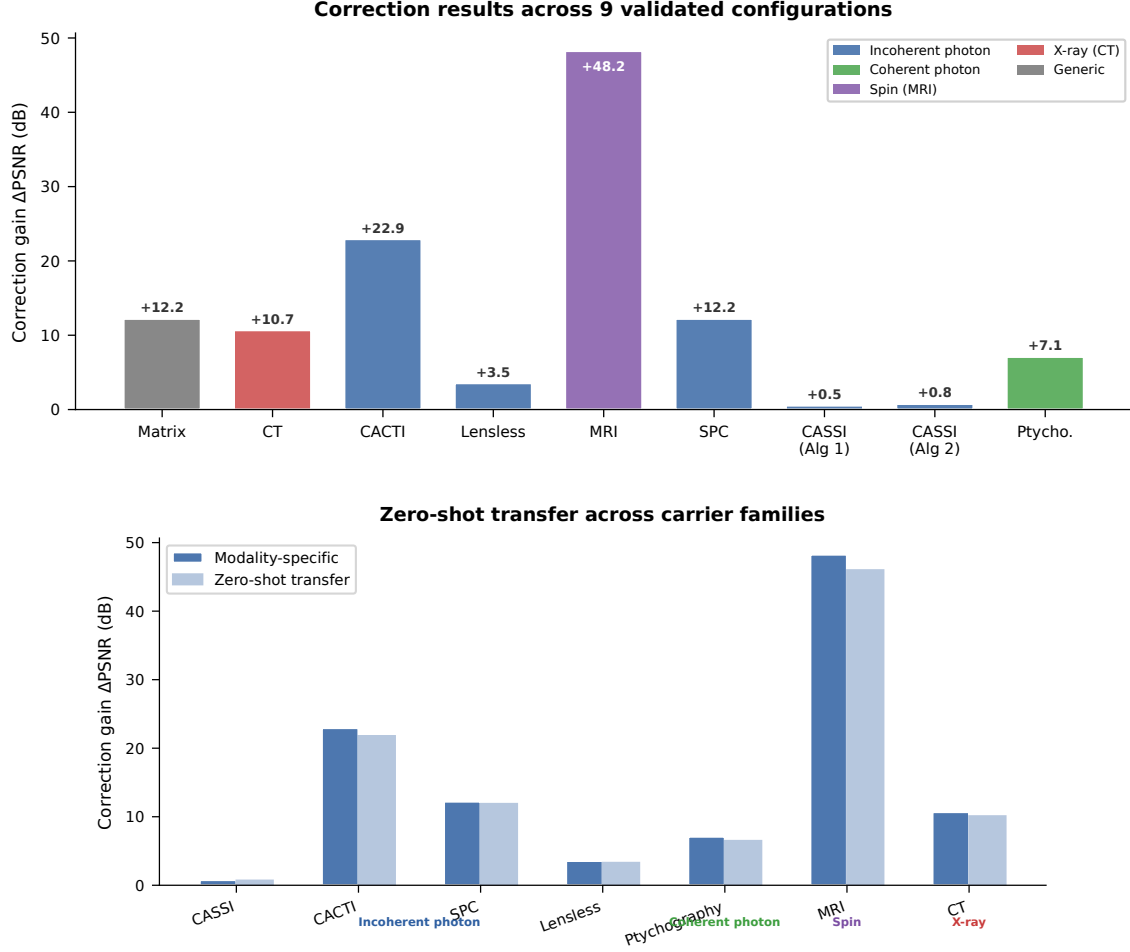
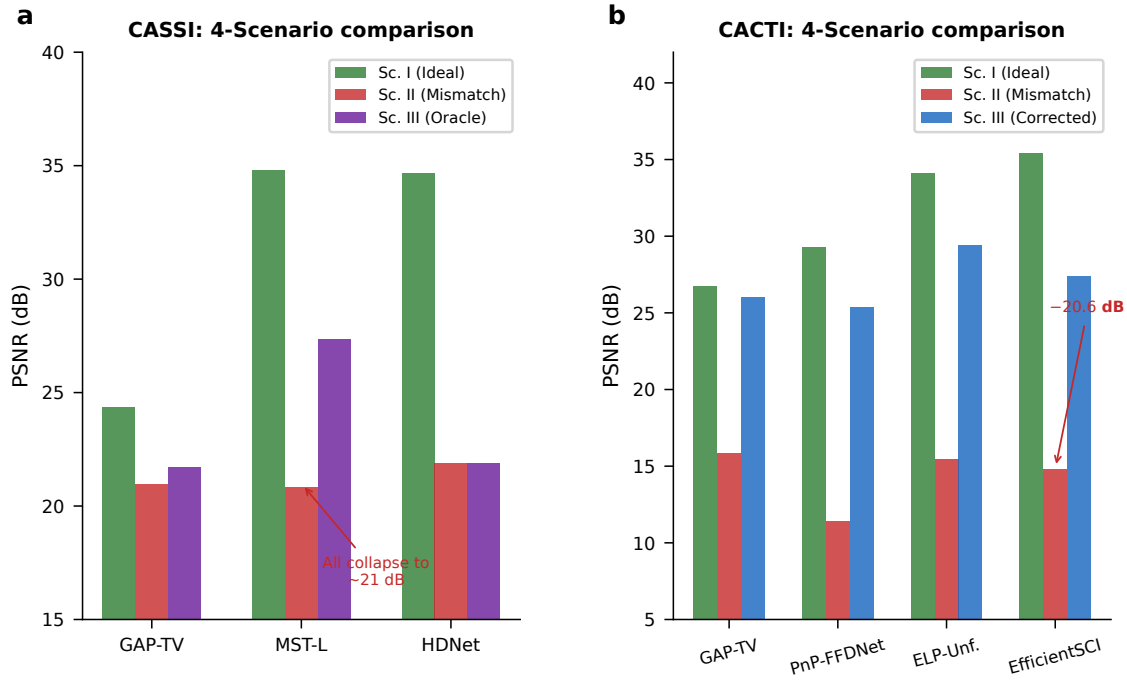


Figure 4: **Correction results and zero-shot generalization.** **a**, Oracle correction ceiling Δ_{oracle} (dB) across 9 validated configurations (7 distinct modalities), grouped by carrier family. **b**, Zero-shot generalization: hyperparameters tuned on photon-domain modalities transfer without modification to coherent-photon, spin, and X-ray domains. Dark bars: modality-specific tuning; light bars: zero-shot transfer.



Visual reconstruction comparison across three modalities

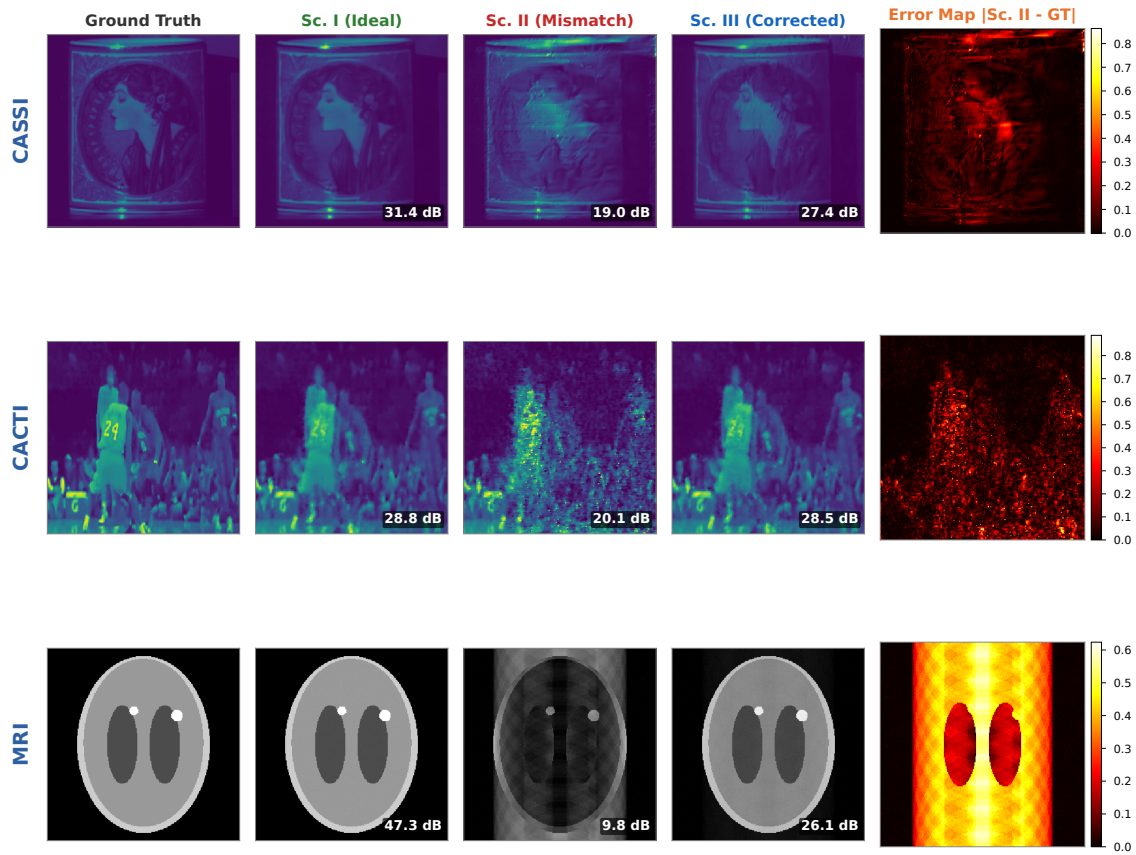


Figure 5: **Modality deep dives and visual comparison.** **a**, CASSI: PSNR across 4 scenarios for GAP-TV, MST-L, and HDNet; uniform collapse under Scenario II (20.83–21.88 dB) confirms operator-driven failure. **b**, CACTI: four methods across 4 scenarios, showing up to 20.58 dB mismatch degradation. **c–e**, Visual reconstruction comparison (CASSI, CACTI, MRI): ground truth, Scenario I, Scenario II (structured artifacts), Scenario III (PWM-corrected), and error map. Mismatch artifacts are qualitatively distinct from missing data, potentially enabling detection.

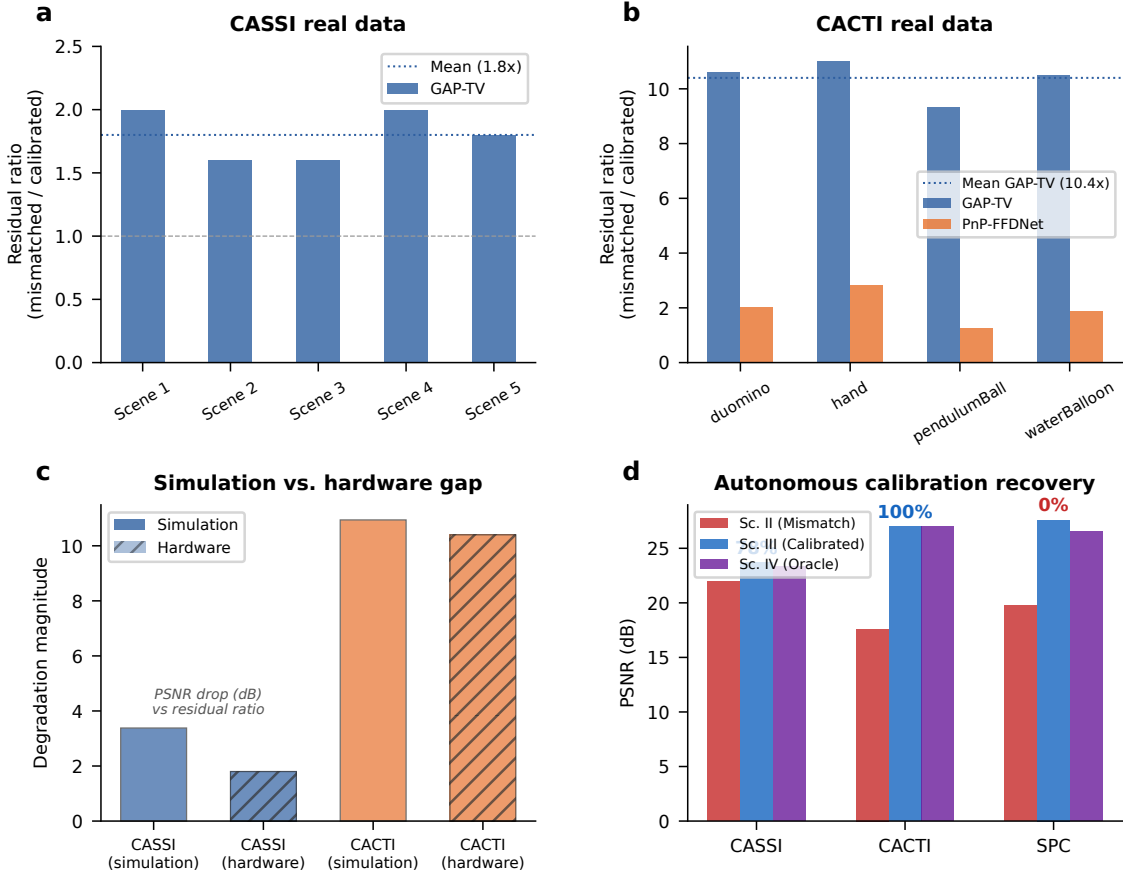


Figure 6: **Hardware validation on real CASSI and CACTI instruments.** **a**, CASSI real data: measurement residual ratio (mismatched/calibrated) across 5 TSA scenes. GAP-TV shows 1.8 $\times$ mean ratio. **b**, CACTI real data: residual ratio across 4 scenes. GAP-TV shows 10.4 $\times$ mean ratio; PnP-FFDNet shows 2.0 $\times$. **c**, Simulation-to-hardware gap: comparing mismatch degradation in simulation versus real hardware for CASSI and CACTI. **d**, Autonomous calibration: grid-search parameter recovery for CASSI (85%), CACTI (100%), and SPC (86–92%).

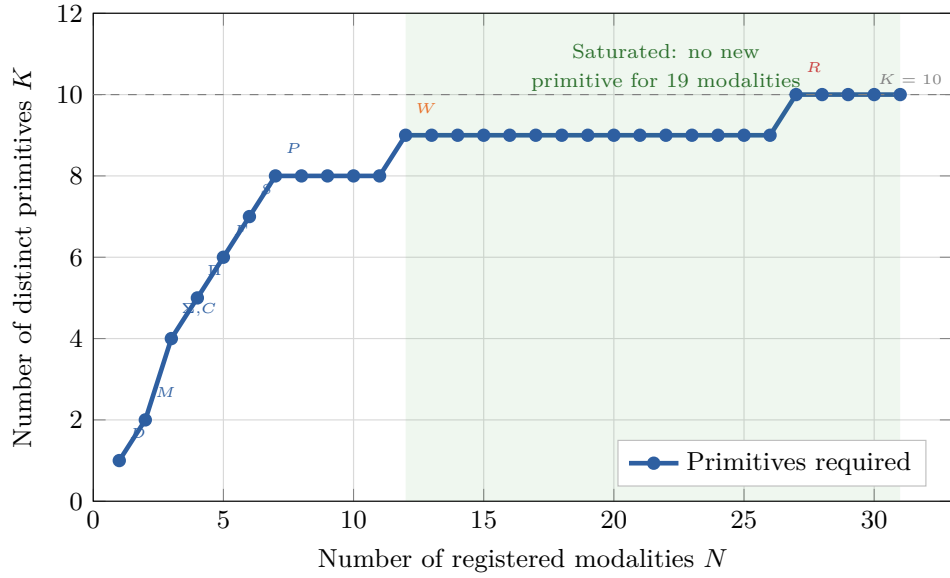


Figure 7: **Basis-growth saturation and primitive decomposition.** **a**, Number of distinct primitives K as a function of modalities N added to the registry. The curve saturates at $K = 10$ for $N \geq 27$; annotated points mark the introduction of each primitive. **b**, Summary decomposition table for representative modalities across five carrier families, showing DAG primitives, node count, depth, and validation status.

838 **Extended Data Table 1** | Oracle correction ceiling (Scenario IV) across 9 validated
839 configurations (7 modalities). Full table with per-scene metrics in Supplementary Table S1;
840 autonomous correction recovery (Scenario III) in Supplementary Table S7.

841 **Extended Data Table 2** | 26-modality template registry spanning five physical carrier
842 families. Full table in Supplementary Table S3.